HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **ETL/ELT data pipelines** using Python for batch data processing, supporting approximately **5 TB** of daily data ingestion.
- Cleaned and validated diverse datasets from **relational databases** (PostgreSQL, Oracle), **CSV**, and **JSON** files, ensuring data integrity for downstream analytics.
- Integrated **RESTful APIs** to ingest external data streams, building lightweight internal services with **FastAPI** that processed ~**1,500 requests per hour**.
- Automated data workflow tasks using prototype **AI agents** and **LLM APIs**, focusing on output quality and logical consistency for reporting purposes.
- Analyzed complex datasets using **pandas**, generating clear **visualizations** and summaries that informed business decisions for operational improvements.
- Built automated **web scrapers** using Python libraries to systematically gather and process unstructured data from target websites, improving data acquisition efficiency by **30%**.
- Containerized data pipeline services using **Docker** and deployed them to **AWS EC2** instances, managing configurations and monitoring resource utilization.

SKILLS

Data Engineering – ETL, ELT, Data Pipelines, Batch Processing, Streaming Processing, Data Warehousing, Relational Databases, SQL, Data Validation, Data Cleansing

Programming Languages – Python, SQL, Bash Scripting

Python Libraries & Frameworks – pandas, NumPy, scikit-learn, FastAPI, Flask, BeautifulSoup, Selenium, Plotly

APIs & Web Services – RESTful APIs, HTTP Methods, JSON

AI/ML Concepts – AI Agents, LLM APIs, Machine Learning, Large Language Models

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins, Linux CLI

PROJECTS

Automated Data Ingestion Pipeline (HCL Technologies)

- Engineered a **Python-based ETL pipeline** to ingest and transform data from disparate sources including relational databases and flat files for a financial-services operations dashboard.
- Implemented robust data validation checks and error handling mechanisms within the pipeline, reducing data quality issues by **22%** for the reporting team.
- Scheduled and monitored pipeline execution using cron jobs and **Linux shell scripting**, ensuring reliable daily data updates.

Customer Data Integration Service (HCL Technologies)

- Developed a **RESTful API** using **FastAPI** to provide an integration point for enterprise customers to push transactional data into a centralized system.
- Designed the API to handle **JSON** payloads and implemented input validation to ensure data consistency, supporting up to **500 concurrent connections** during peak loads.
- Wrote comprehensive unit tests using pytest, achieving over **90% code coverage** for the service's core logic.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)